OPEN AGENT HACKATHON 2026 • TEAM: LIGHT OF NUMBERS

MarginWhy: Multi-Agent Root-Cause Investigation

An Adversarial Multi-Agent Platform Explaining Retail Profit Margin Drops & Cross-Silo Operational Disruptions with Mathematical Provenance.

🎯 **Primary Track**

Track 02

"The Agent That Can Explain Why" — Cross-examines conflicting hypotheses and delivers citation-backed root causes.

⚡ **In-Kind Sponsor Target**

NVIDIA Nemotron

100% reasoning routed via Meterless Nightshift to `nvidia/nemotron-3-super-120b-a12b` (9,400+ Nemotron trace events).

🛠️ **Tooling & Architecture**

Cursor + Meterless H-MEM

Built with Cursor Composer agentic workflows; Powered by Meterless 3-tier H-MEM & Zetaris Lightning SQL.

✓ 190 / 190 Tests Passed (100%) ✓ Section 5.1 Matrix: 5/5 PASS ✓ 13,051 Trace Events ✓ Sub-Second Execution (0.77s)

SECTION 1 • PROBLEM & PERSONA

The Problem: Data Silos Obscure the Real "Why"

When net profit margin collapses, operational leaders face fragmented telemetry across isolated enterprise systems.

👤 Target User Persona

Retail Operations & Merchandising Leads

Responsible for omnichannel P&L, inventory allocations, fulfillment SLAs, and margin protection across thousands of retail items.

The Pain Point: Surface metrics (like return rates) trigger knee-jerk blame, while cross-silo failures (inventory stockouts + delivery transit surcharges) remain invisible.

💥 The Anomaly (Electronics Q3)

-18.5% Net Margin

- Initial bias: Executives assumed defective product returns (H1).
- Data reality: National returns rose only 1.2% ($103K impact).
- True root cause: Northeast DC stockout (84% drop) + urban freight delays (34% transit hike) triggered 2,410 cancellations and $39.50 emergency courier surcharges ($1.78M impact).

SECTION 2 • SOLUTION ARCHITECTURE

MarginWhy: Adversarial Multi-Agent System

Moving beyond naive linear pipelines to an iterative arbitration loop with a dedicated Skeptical Critic.

🚫 Why Naive Chains Fail

Linear architectures (`Planner -> Retriever -> Writer`) blindly hallucinate or accept the first correlation they encounter.

In retail analytics, this leads to fatal business errors: cutting marketing or firing suppliers when logistics congestion was the true culprit.

🛡️ The MarginWhy Advantage

- **Arbitration Authority:** SkepticalCritic possesses the power to issue `CLAIM_BLOCKED` events and force orthogonal re-branching.
- **Audit Ledger:** Every hypothesis mutation is recorded in an immutable trust ledger.
- **Actionable Directives:** Produces prioritized operational remedies with dollar recovery estimates.

SECTION 3 • ITERATIVE MULTI-AGENT LOOP

4 Specialized Collaborative Agents

Coordinated by MarginWhyOrchestrator across strict Markovian state boundaries.

1. Planner

HypothesisPlanner

Decomposes questions into testable hypotheses (H1, H2, H3). Reads rejected list from H-MEM so it never repeats failed paths.

2. Retriever

ZetarisRetriever

Formulates Lightning SQL with `LIMIT` and 30s timeout. Joins cross-silo tables (TPCDS_DB + NY_TAXI) with automatic type coercion.

3. Critic

SkepticalCritic

Rigorous auditor. Scores statistical sufficiency. If confidence < 0.70, emits `CLAIM_BLOCKED` and triggers `BRANCH_RETRY`.

4. Synthesizer

ActionReportWriter

Compiles ONLY verified claims into structured executive reports with exact SQL citations and explicit data limitations.

SECTION 4 • TECHNICAL STACK & 190-TEST RIGOR

Production-Grade, Zero-Friction Architecture

Clean separation of concerns with sub-60-second execution on standard CPU hardware.

Backend & Serving

- **FastAPI + ASGI:** Async serverless endpoints & REST API.
- **Pydantic v2:** Strict validation of queries & scenarios.
- **Docker & Vercel:** 1-command container and cloud deployment.
- **190 Automated Tests:** 100% pass across H-MEM conformance, security & adversarial test suites.

Data & Intelligence

- **Zetaris Lightning SQL:** Cross-database federated querying.
- **Meterless H-MEM:** 3-tier memory engine with trust ledger.
- **NVIDIA Nemotron:** `nemotron-3-super-120b-a12b` inference.
- **13,051 Trace Events:** Over 9,400 events stamped with Nemotron model tags.

Frontend Experience

- **Executive Bento Grid:** Obsidian dark theme with glassmorphism.
- **SVG Waterfall Chart:** Dynamic financial bridge visualization.
- **Interactive Simulator:** Real-time margin recovery sliders.
- **1-Click Audit Modal:** Live Section 5.1 verification matrix (5/5 PASS).

SECTION 5 • SPONSOR TECHNOLOGY INTEGRATION

Load-Bearing Sponsor Tech & Uniqueness

Deep, functional integration satisfying 100% of Hackathon technical mandates.

Zetaris AI Data Harness

• Unifies `TPCDS_DB` (inventory, web_returns, web_sales) and `NY_TAXI` (transit delays, surcharges) into a single logical SQL interface.
• Handles scientific string coercion and upstream timeout resilience (30s timeout + automatic sample fallback).

Meterless H-MEM & Nightshift

• **H-MEM:** Implements 3 tiers (`short_term`, `working`, `long_term`) with Markovian carryover and sleep-cycle consolidation.
• **Nightshift:** Directs all inference calls to NVIDIA Nemotron, logging `model: "nvidia/nemotron-3-super-120b-a12b"` on every JSONL trace event.

✨ What Was Unique?

• **Adversarial Critic Veto:** Authority to block shallow claims (`CLAIM_BLOCKED`) prevents hallucinated conclusions.
• **Cryptographic Trust Ledger:** Immutable SHA-256 audit trail ensures forensic explainability.
• **Cross-Silo Bridge:** Connects e-commerce sales with urban mobility delays.

SECTION 6 • PRODUCT VISUALS & MULTI-SCENARIO UX

Executive Command Console & Simulator

Designed for immediate clarity, interactive scenario testing, and live verification.

📊 Pure SVG Waterfall

Visualizes the margin reconciliation bridge: Base Margin (22.5%) -> Refuted Returns (+0.0%) -> DC Stockouts (-12.2%) -> Freight Surcharges (-6.3%) -> Final Depressed Margin (4.0%).

🎛️ What-If Simulator

Interactive recovery sliders allow executives to simulate inventory reallocation and off-peak courier routing, calculating projected dollar recovery (+$1.28M) in real time.

🔍 Live Submission Audit

Built-in audit modal runs `scripts/verify_submission.py` live on demand, displaying passing verification across all 5 Section 5.1 Hackathon criteria in 2.1 seconds.

SECTION 7 • FUTURE ENHANCEMENTS

Production Roadmap & Next Horizons

Scaling MarginWhy from hackathon prototype to enterprise enterprise supply-chain intelligence.

📡 Real-Time IoT Telemetry

Integrate live RFID and telematics streams from warehouse docks and transit couriers into Zetaris for sub-second anomaly detection.

🤖 Autonomous Remediation

Connect ActionReportWriter directly to ERP/WMS systems (SAP, Oracle) to automatically draft emergency purchase orders and reroute carriers.

🎲 Monte Carlo Risk Engine

Pair Meterless World Model engine with Nemotron inference to simulate thousands of disruption scenarios and stress-test gross margins against port strikes.